

**REPORT ON
PROJECT BASED LEARNING**

Carried out on

Dermatology Disease Classification Using RBF Submitted to

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(Off-Campus Centre, Nitte Deemed to be University, Nitte – 574 110, Karnataka, India)

In partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology

In

Robotics & Artificial Intelligence By

Aashirvad A. Poojary

Bindu R.

NNM23RI010

NNM23RI13

Khushi K.

Vedant Mahalle

NNM23RI030

NNM23RI031

Under the guidance of DR.

RASHMI P. SHETTY

Associate Professor

Department of Robotics & Artificial Intelligence



**NMAM INSTITUTE
OF TECHNOLOGY**

CONTENT

No.	Title	Page No.
	Abstract	3
1	Introduction	3
2	Literature Review	4-5
3	Dataset Description	5-6
4	Methodology	6-7
5	Model Architecture	7
6	Results and Discussion	7-8
7	Conclusion	10-11
8	References	11-12

Abstract

Dermatological diseases are among the most common medical conditions worldwide, and their accurate diagnosis is crucial for effective treatment. Due to the similarity of symptoms among different skin diseases, manual diagnosis can sometimes be difficult and time-consuming. Machine learning techniques provide an opportunity to develop automated diagnostic support systems that assist dermatologists in making accurate decisions.

In this project, a dermatological disease classification system based on a Radial Basis Function (RBF) model is proposed. The dermatology dataset containing 34 clinical attributes and six disease classes was used for training and evaluation. The methodology includes data preprocessing, correlation analysis, feature scaling, clustering using K-Means, RBF transformation, and classification using machine learning algorithms such as Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest for comparison. Experimental results show that the tuned RBF model achieves high classification accuracy and demonstrates the effectiveness of similarity-based learning for dermatological diagnosis.

1. Introduction

Skin diseases affect millions of people worldwide and can significantly impact quality of life. Dermatological conditions such as psoriasis, seborrheic dermatitis, and lichen planus often exhibit similar symptoms like redness, itching, and inflammation, which makes diagnosis challenging. Traditionally, dermatologists rely on clinical experience and laboratory tests to determine the correct disease category.

With advancements in artificial intelligence, machine learning models can assist medical professionals by analyzing clinical data and identifying hidden patterns. These systems can process multiple patient attributes simultaneously and provide reliable classification results.

In this project, an RBF-based machine learning model is developed to classify dermatological diseases. Unlike traditional neural networks that rely on weight optimization, the RBF model uses similarity measures between patient records and representative patterns. By combining clustering techniques with nonlinear classifiers, the proposed system aims to improve diagnostic accuracy and support dermatologists in decision making.

The objective of this project is to design and implement an automated dermatological disease classification system using a Multi-Layer Perceptron (MLP) model. The system utilizes structured clinical data to classify patients into different dermatological disease categories. MLP is chosen due to its ability to learn non-linear relationships in tabular medical data. The proposed system aims to support dermatologists by providing reliable classification results and reducing diagnostic effort.

2. Literature Review

1) Radial Basis Function (RBF) Neural Network Classification based on Consistency Evaluation Measure

The paper titled “Radial Basis Function Neural Network Classification based on Consistency Evaluation Measure” by Aye Mya Thandar and Myo Kay Khaing discusses the use of an RBF neural network for classifying medical datasets. In their work, the authors used four medical datasets obtained from the UCI Machine Learning Repository. The RBF neural network used in the study consists of three layers: input layer, hidden layer, and output layer. The hidden layer applies radial basis functions to process the input data and help in performing the final classification. The datasets used in the study are Breast Cancer, Lymphography, Sick Thyroid, and Hepatitis. The Breast Cancer dataset contains 699 instances and 9 attributes, and it is mainly used to classify tumors as benign or malignant. Before performing feature selection, the ID number attribute was removed since it does not contribute to classification. The Lymphography dataset contains 148 instances with 18 attributes and has four classes. The Sick Thyroid dataset includes 3163 instances and 30 attributes with two classes. In this dataset, the age and steroids attributes were removed because they were not useful for classification. The Hepatitis dataset contains 295 instances and 20 attributes, where 155 samples were used for training and 140 samples were used for testing. After applying the proposed method, the dimensionality of the datasets was reduced, which helped in speeding up the training process and improving classification accuracy. The results showed accuracy values of 84.10% for Hepatitis, 93.92% for Sick Thyroid, 79.46% for Lymphography, and 85.19% for the Breast Cancer dataset. From these results, it can be understood that combining RBF neural networks with consistency-based feature selection improves classification performance while also reducing computational complexity.

2) Classification of Skin Cancers using Radial basis Function Network

Computer-aided diagnostic (CAD) systems have been developed to assist in the early detection and classification of skin cancer. In this study, a Radial Basis Function Network (RBFN) is used to classify skin lesions as benign or malignant melanoma. The process begins with image preprocessing, where median filtering is applied to remove noise such as thin hairs and artifacts. After preprocessing, segmentation is performed to isolate the lesion area from the surrounding skin. Then, 29 features related to shape, color, and texture are extracted from the lesion image and used as input for the classifier. The RBFN model analyzes these features and performs the classification. The proposed system achieved 94.44% accuracy, 100% sensitivity, and 90.91% specificity, showing good performance in skin cancer detection. However, the study has some limitations, such as the dataset size not being clearly mentioned and the number of principal components retained not being clearly explained.

3) IDENTIFICATION OF ERYTHEMATO-SQUAMOUS SKIN DISEASES USING EXTREME LEARNING MACHINE AND ARTIFICIAL NEURAL NETWORK

Sunday Olusanya Olatunji and Hossain Arif (2013) Compared Artificial Neural Network (ANN) and Extreme Learning Machine (ELM) for classification of Erythematous-Squamous Diseases of the Skin. In the preprocessing stage, the attribute of age was excluded from the data.

For the ANN, the architecture of the model employed is a single hidden layer neural network, back-propagation learning algorithm, sigmoid and linear activation function, learning rate of 0.01, and 1000 epochs of training. On the other hand, for the ELM, the architecture of the model employed is a single hidden layer feed-forward network, 500 hidden neurons, and a Hardlim activation function, where the hidden layer weights are randomly selected, and output layer weights are calculated analytically.

The results of both models are obtained for 80-20, 70-30, 60-40, and 50-50 split values of data for testing and training, where ANN results are 82.74%, 79.17%, 77.26%, and 78.09%, while ELM results are 94.5%, 98.17%, 96.58%, and 98.36%

for the corresponding values of data split for testing, proving ELM to be much better than ANN in terms of accuracy and learning speed.

4) Erythemato Squamous Disease Diagram Using SVM and RBF Neural Network

This Research was done by Vojislav Kecman (Virginia Commonwealth University, USA) and Mirna Kikec (Emergency Medical Institution Zagreb, Croatia). Their idea was basically automating diagnostic system that classifies skin diseases that are difficult to classify. They faced problem that many diseases share similar symptoms so manual diagnosis is hard. As a solution they used SVM and RBF to classify diseases. They used SVM because dataset was small and SVM works extremely powerful on small and non-linear medical dataset. They divided dataset into 2 categories:

- i) Clinical Features (11) – without microscope
- ii) Histopathological Features (22) – with microscope. They divided dataset into 90 % training and 10 % testing data. For RBF they got accuracy of 98 % and for SVM they got accuracy of 100% (sometimes 100 % accuracy is considered as model is overfitting) but when they tested model on unseen data it performed well

3. Dataset Description

The dataset used in this project is a dermatology dataset containing structured clinical records of patients diagnosed with various skin diseases. Each instance in the dataset represents a single patient record.

The dataset consists of:

- **Number of instances:** 366
- **Number of input features:** 34
- **Type of data:** Tabular, numerical
- **Output type:** Multi-class categorical label

The 34 input features describe clinical symptoms and observations commonly used by dermatologists during diagnosis. These attributes include features related to erythema, scaling, papules, plaques, itching, and other dermatological indicators. Since these symptoms often co-occur in real medical conditions, some features exhibit correlation.

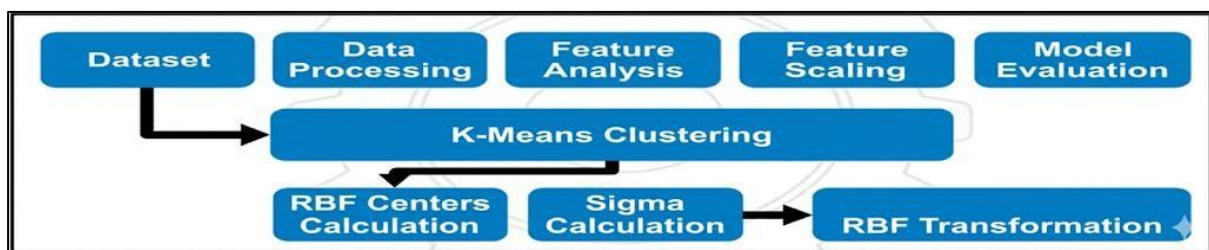
The target variable represents the diagnosed dermatological disease and is classified into six distinct classes:

1. **Psoriasis** – A chronic autoimmune skin disease characterized by red, scaly patches and inflammation.
2. **Seborrheic Dermatitis** – A condition causing scaly patches and red skin, commonly affecting areas rich in oil glands.
3. **Lichen Planus** – An inflammatory disease presenting as purplish, itchy, flat-topped lesions.
4. **Pityriasis Rosea** – A self-limiting skin rash that usually begins with a single large patch followed by smaller lesions.
5. **Chronic Dermatitis** – Long-term skin inflammation associated with itching, redness, and irritation.
6. **Pityriasis Rubra Pilaris** – A rare skin disorder characterized by reddish-orange scaling patches and thickened skin.

Since each patient belongs to exactly one disease category, the dataset represents a multi-class classification problem, making it suitable for neural network-based models such as MLP.

4. Methodology

The proposed system follows a sequence of steps to ensure reliable classification.



4.1 Data Preprocessing

The dataset was first analyzed to identify missing values and inconsistencies. Missing values were replaced using **median imputation**, which maintains the central tendency of the data and prevents errors during training.

Figure 2: Correlation matrix of dermatological dataset features.

4.2 Correlation Analysis

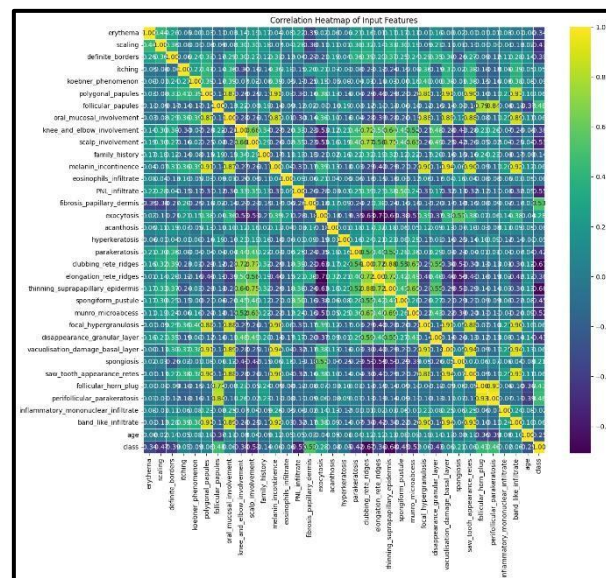


Figure 2: Correlation matrix of dermatological dataset features.

Explanation: -

A correlation matrix was computed to analyze the relationships among the dermatological features. The heatmap visualization helps identify strong positive or negative correlations between symptoms. Highly correlated attributes indicate that certain dermatological symptoms frequently occur together in specific diseases.

4.3 Graph Analysis

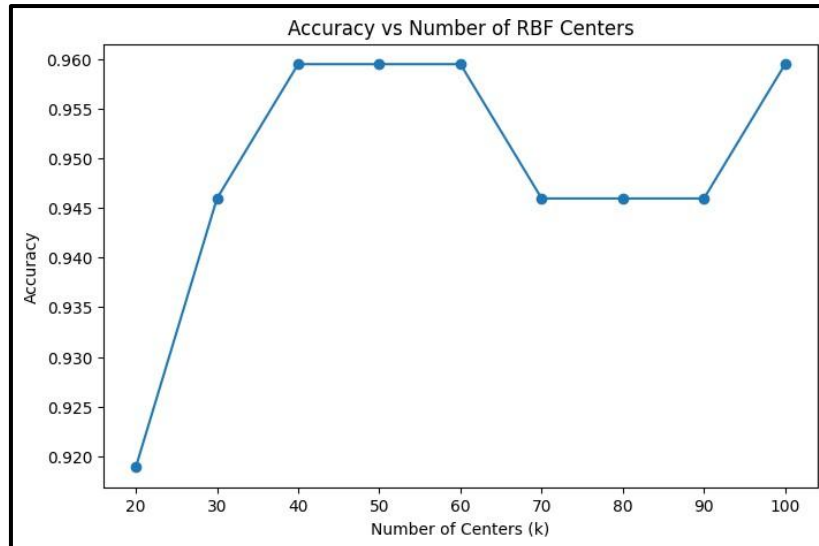


Figure 3: Accuracy VS Number of RBF Centers

Explanation: -

The graph illustrates the relationship between the number of RBF centers and classification accuracy. Increasing the number of centers improves the model's ability to capture complex patterns in the dataset. However, very large values may increase computational cost without significant accuracy improvement.

5. Model Architecture

The architecture of the RBF-based classification system consists of three layers.

5.1 Input Layer

The input layer contains 34 features, representing dermatological symptoms observed during patient examination.

5.2 RBF Layer

The hidden layer consists of k radial basis neurons, where each neuron corresponds to a cluster center obtained from K-Means clustering.

Each neuron computes the similarity between the input and its corresponding center.

5.3 Output Layer

The transformed features are then passed to a classification algorithm that predicts the final disease category.

6. Results and Discussion

6.1 Accuracy

The performance of the proposed RBF-based system was evaluated using classification accuracy. The final model achieved an overall accuracy of approximately: 97.30%

6.2 Key Observations from k Value Tuning

Different values of k (number of cluster centers) were tested

k Values	Accuracy	Observation
20	91.89%	Limited Pattern Representation
30,40,50,60	95.95%	Accuracy Improved and Stable
70,80	94.59%	Accuracy Dropped/ Possible overfitting
90	97.30 %	Best performance; optimal balance between representation and generalization
100	95.95%	No further improvement; additional centers increased complexity

Table: Key Observation

6.3 Comparison with other Algorithms

The algorithm comparison graph shows the accuracy of different machine learning models used in the project. The RBF-SVM model achieved the highest accuracy, indicating its strong ability to capture complex patterns in the dermatology dataset. While Random Forest also performed well, the KNN model showed comparatively lower accuracy, highlighting that nonlinear models such as RBF-SVM are more suitable for this classification problem.

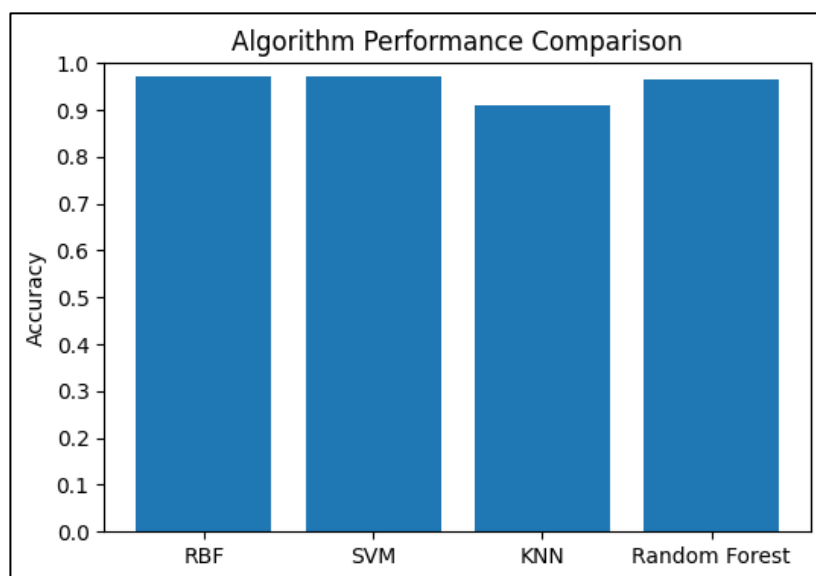


Figure 3: Algorithm Performance

Comparison Table

Model Names	Accuracy
SVM	97.30 %
Random Forest	95.95 %
KNN	91.89 %

6.4 Precision, Recall, and F1-Score

In addition to accuracy, the model performance was evaluated using precision, recall, and F1-score to ensure balanced classification across all disease classes. The model achieved a weighted precision of 0.975, recall of 0.973, and F1-score of 0.974, indicating strong and balanced prediction performance. These metrics confirm that the model effectively minimizes both false positives and false negatives, which is important for reliable medical diagnosis.

Metric	Value
Accuracy	97.30%

Precision	0.975
Recall	0.973
F1-score	0.974

Table : Performance Metrics of the Proposed RBF Model

6.3 Confusion Matrix Analysis

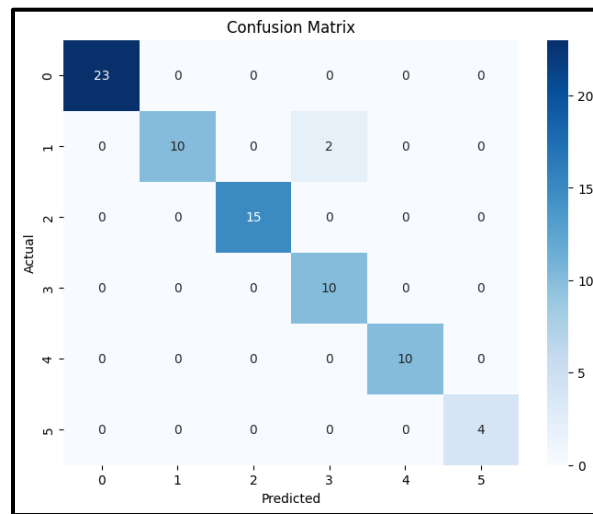


Figure 2: Confusion matrix of the MLP classifier on the test dataset.

Explanation: -

Figure 2 presents the confusion matrix obtained for the RBF classifier on the test dataset. The diagonal elements of the matrix represent correctly classified instances, while the off-diagonal elements indicate misclassifications.

Most samples are correctly classified, as evidenced by the strong diagonal values. Minor misclassifications occur between disease classes with similar clinical features, which is expected due to overlapping symptom characteristics in dermatological conditions.

7. Conclusion

This project successfully demonstrates the application of a Multi-Layer Perceptron (MLP) model for the classification of dermatological diseases using clinical data. The proposed system effectively captures the non-linear relationships between dermatological features and disease classes, resulting in high classification accuracy. Careful data preprocessing, correlation analysis, and feature scaling played a crucial role in improving the learning stability and performance of the MLP model.

Despite the encouraging results, the performance of the system is influenced by the quality and size of the dataset. Certain disease classes with similar clinical characteristics occasionally lead to misclassification, highlighting the inherent complexity of dermatological diagnosis.

Future improvements to this work include experimenting with alternative machine learning models such as Random Forest and Support Vector Machines for comparative analysis. Further performance gains may also be achieved through systematic hyperparameter tuning and by training the model on larger and more diverse datasets. These enhancements can improve generalization and make the system more robust for real-world clinical applications.

8. References

1. ***Erythematous-Squamous Diseases Diagnosis by Support Vector Machines and RBF Neural Networks***
Authors: Vojislav Kecman, Mirna Kikec
ICAISC 2010 – International Conference on Artificial Intelligence and Soft Computing
https://link.springer.com/chapter/10.1007/978-3-642-13208-7_74
2. **Radial Basis Function (RBF) Neural Network Classification based on Consistency Evaluation Measure**
Authors: Aye Mya Thandar, Myo Kay Khain
<https://research.ijcaonline.org/volume54/number15/pxc3882463.pdf>

